# Measures of Central Tendency (1)

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# Descriptive measures

- In previous chapter, we discussed how a raw data set can be organized and summarized by tables and graphs.

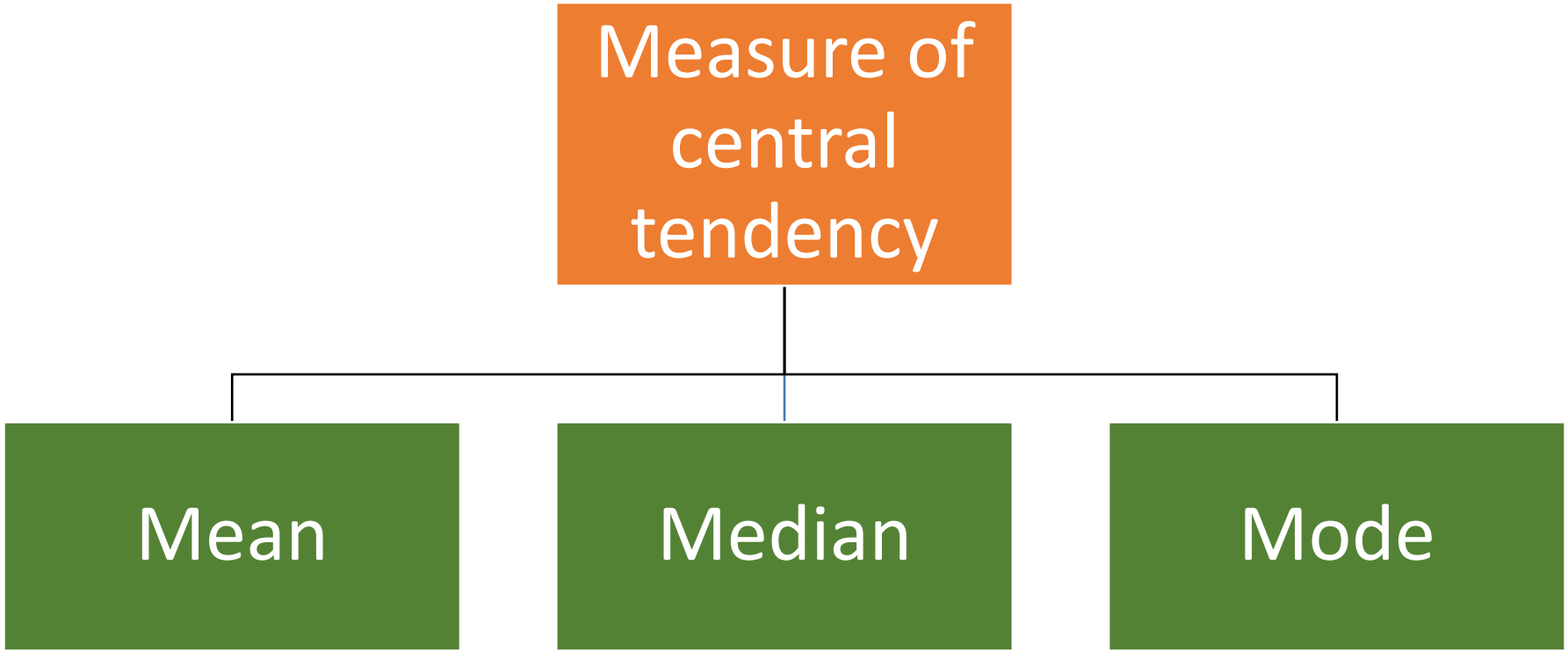
- Another method of summarizing data set precisely is to compute number (a single number).

- Number that can be describe data sets are called descriptive measures

# Measures of Central Tendency

- It is a summary measure
- Attempts to describe the whole data set with a single value
- Represents the center of the distribution

# Measures of Central Tendency



# Measures of Central Tendency

**Types of mean :**

1. Arithmetic mean
2. Geometric mean
3. Harmonic mean

# Arithmetic Mean (AM)

- It is simply the sum of the observations divided by the total number of observations.

- For example: Consider the values 6,8,4,7,2,9

- $\boldsymbol{Mean = \frac{(6+8+4+7+2+9)}{6} = 6\ Unit}$

# Formula: AM

| Ungrouped Data | Grouped Data |
| --- | --- |
| Let $x_1, x_2, \ldots, x_n$ are some observations.<br><br>Arithmetic mean for ungrouped data can be written as,<br><br>$$\bar{X}_{AM} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum x_i}{n}$$<br><br>For example: Consider the values: 6,8,4,7,2,9<br><br>$$\boldsymbol{Mean} = \frac{(\mathbf{6+8+4+7+2+9})}{\mathbf{6}}$$<br>$$= \mathbf{6}\ \boldsymbol{Unit}$$ | One additional term "$f_i$", frequency.<br><br>Arithmetic mean for grouped data can be written as,<br><br>$$\bar{X}_{AM} = \frac{\sum(f_i \times x_i)}{\sum f_i}$$<br><br>Here,<br>$f_i = Frequency$;<br>$x_i = Class\ mid - value$ |

# Group data: AM

| Class | Frequency |
|---|---|
| 2-6 | 2 |
| 6-10 | 6 |
| 10-14 | 4 |
| 14-18 | 2 |

**Calculate mean or arithmetic mean or average from this table.**

# Group data: AM

$f_i = Frequency$
$x_i = Mid\ value$

| Class | Frequency |
|---|---|
| 2-6 | 2 |
| 6-10 | 6 |
| 10-14 | 4 |
| 14-18 | 2 |

**Calculate mean or arithmetic mean or average from this table…**

$$\bar{X}_{AM} = \frac{\sum(f_i \times x_i)}{\sum f_i}$$

# Group data: AM

| **Class** | $x_i$ | $f_i$ | $f_i \times x_i$ |
|---|---|---|---|
| 2-6 | 4 | 2 | 8 |
| 6-10 | 8 | 6 | 48 |
| 10-14 | 12 | 4 | 48 |
| 14-18 | 16 | 2 | 32 |
| **Total** | | 14 | 136 |

**Calculate mean or arithmetic mean or average from this table…**

$$\bar{X}_{AM} = \frac{\sum(f_i \times x_i)}{\sum f_i} = \frac{136}{14} = 9.71$$

# Class work

| Age (in Years) | Frequency |
|---|---|
| 10-20 | 5 |
| 20-30 | 9 |
| 30-40 | 5 |
| 40-50 | 3 |
| 50-60 | 6 |
| 60-70 | 2 |

**Calculate mean or arithmetic mean or average from this table.**

# Group data: AM

| Class | Frequency |
|---|---|
| 2-6 | 2 |
| 6-10 | 6 |
| 10-14 | $f_1$ |
| 14-18 | 2 |

**Find the value of $f_1$, when the average in 9.54.**

# Group data: AM

| Class | $x_i$ | $f_i$ | $f_i \times x_i$ |
|---|---|---|---|
| 2-6 | 4 | 2 | 8 |
| 6-10 | 8 | 6 | 48 |
| 10-14 | 12 | $f_1$ | $12\ f_1$ |
| 14-18 | 16 | 2 | 32 |
| **Total** | | $10 + f_1$ | $88 + 12f_1$ |

$$\bar{X}_{AM} = \frac{\sum(f_i \times x_i)}{\sum f_i}$$

$$9.54 = \frac{88 + 12f_1}{10 + f_1}$$

$f_1$ =??

# Class work

| Age (in Years) | Frequency |
|---|---|
| 10-20 | 6 |
| 20-30 | 8 |
| 30-40 | $f_1$ |
| 40-50 | 3 |
| 50-60 | $f_2$ |
| 60-70 | 2 |

**Find the values of $f_1$ 1and $f_1$, when the average is 31.54, and total number of observations is 26.**

# Geometric Mean

- Geometric mean is useful when the data is in geometric progression
- Exponential growth, or
- Growth over the year/ change over a period of times, or
- Observations are ratios

# Geometric Mean

GM for ungrouped data:

$$\bar{x}_{\mathrm{GM}} = (x_1 \times x_2 \times \cdots \times x_n)^{\frac{1}{n}}$$

GM for grouped data:

$$\bar{x}_{\mathrm{GM}} = \left(x_1^{f_1} \times x_2^{f_2} \times \cdots \times x_n^{f_n}\right)^{\frac{1}{\sum f_i}}$$

# Geometric Mean

- For example: 6,8,4,7,2,9

- $\bar{X}_{GM} = (6 \times 8 \times 4 \times 7 \times 2 \times 9)^{\frac{1}{6}} = 5.37$

# Geometric Mean

- Find the geometric mean for the following distribution:

| Class | Frequency |
|---|---|
| 2-6 | 2 |
| 6-10 | 6 |
| 10-14 | 4 |
| 14-18 | 2 |

# Geometric Mean

- If $x_i$ are the percent value (rate) for a given time t,

$$Average\ growth\ rate = \{(1 + r_1) \times (1 + r_2) \times \cdots \times (1 + r_n)\}^{\frac{1}{n}} - 1$$

$$Average\ depreciation\ rate = \{(1 - r_1) \times (1 - r_2) \times \cdots \times (1 - r_n)\}^{\frac{1}{n}} - 1$$

# Geometric Mean

- Suppose the value of an assets grew by 5% in the first year, 10% in the second year, 8% in the third year. What was the average growth rate for these three years.

- Solution: Since, this is a geometric progression rate. So, geometric mean may used here.

Average growth rate, $\bar{X}_{GM} = \{(1 + 0.05) \times (1 + 0.10) \times (1 + 0.08)\}^{\frac{1}{3}} - 1$

$$\bar{X}_{GM} = 0.077$$

Comment: Thus, the average growth rate is 7.7%

# Geometric Mean

- Let's consider an investment that depreciated by 5% in the first year, 6% in the second year, and 10% in the third year. What was the average depreciation rate over these three years?

$$Average\ depreciation\ rate = \{(1 - r_1) \times (1 - r_2) \times \cdots \times (1 - r_n)\}^{\frac{1}{n}} - 1$$

# Harmonic Mean

- Harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of the individual values.
- It is necessary to compute the average of some variables such as the average speed, average velocity, and so on.
- When dealing with rates, harmonic mean is more appropriate

# Harmonic Mean

HM for ungrouped data:

$$\bar{X} = \frac{\text{n}}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$$

HM for grouped data:

$$\bar{X} = \frac{\sum f_i}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \cdots + \frac{f_n}{x_n}}$$

# Harmonic Mean

- For example: 6,8,4,7,2,9
- $\bar{X}_{HM} = \frac{6}{\frac{1}{6}+\frac{1}{8}+\frac{1}{4}+\frac{1}{7}+\frac{1}{2}+\frac{1}{9}} = 4.63$

# Harmonic Mean

- A truck travels 40 miles at 60 mph, 60 miles at 50 mph, and 50 miles at 80 mph. What is the average speed of the trip?

- Solution: Here, Distance: 40 miles, 60 miles, 50 miles

Speed: 60 mph, 50 mph, 80 mph

The mean (harmonic mean) can be written as,

$$HM = \frac{\sum f_i}{\sum \left(\frac{f_i}{x_i}\right)} = \frac{40 + 60 + 50}{\frac{40}{60} + \frac{60}{50} + \frac{50}{80}} = 60.02\ mph$$

# Some points…

- $AM \geq GM \geq HM$
- $GM = \sqrt{AM \times HM}$

# Combined Mean

- If $\bar{X}_1$ and $\bar{X}_2$ are the means with respective numbers of observations $n_1$ and $n_2$ of two data sets expressed in the same measuring units, then combined mean is given by,

$$\bar{X}_c = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$$

# Combined Mean

- Consider 2 section of any University. A random sample A of 15 students has mean CGPA 3.65, and another random sample B of 10 employees has mean CGPA 3.57. Compute the arithmetic mean of the combined sample.

- Solution:

$$\bar{X}_c = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2} = 3.618$$

# Weighted mean

- The weighted mean is a special case of the arithmetic mean. It occurs when there are several observations of the same value.

- To explain- Suppose a shop offers three different kinds of ice-cream vanilla, Chocolate and Strawberry for Tk. 100, Tk. 120 and Tk. 140. Of the last 10 ice-cream sold 3 were vanilla, 4 were Chocolate and 3 were Strawberry. To find the mean price of the last 10 ice-cream sold we can calculate using the usual formula of the arithmetic mean as follows

$$\bar{X}_{AM} = \frac{(100 + 100 + 100 + 120 + 120 + 120 + 120 + 140 + 140 + 140)}{10} = 120$$

# Weighted mean

$$\bar{X}_{WM} = \frac{\sum(w_i \times x_i)}{\sum w_i}$$

- An easier way to find the mean selling price is to determine the weighted mean.

$$\bar{X}_{WM} = \frac{(3 \times 100) + (4 \times 120) + (3 \times 140)}{10} = 120$$

# Class Work

- Madina Construction Company pays its part time employees hourly basis. For different level of employee, the hourly rate is Tk 100, Tk 125 and Tk 150. There are 260 hourly employees, 140 of which are paid at Tk 100 rate, 100 at Tk 125 and 20 at the Tk 150 rate What is the mean hourly rate paid to the employees?

# Thank You!